Placement Predictor using Random Forest Optimization

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1. Introduction

In modern academic institutions, early identification of students at risk of not being placed is critical for targeted intervention strategies. This study focuses on building a predictive model to classify students as “at-risk” using machine learning techniques. A **Random Forest Classifier** is employed due to its robustness, ability to handle nonlinear relationships, and resistance to overfitting.

The primary objective is not only to build a predictive model but also to evaluate different **hyperparameter optimization strategies**, specifically **Grid Search** and **Randomized Search**, while analyzing the impact of evaluation metrics such as **Accuracy** and **F1-Score**.

2. Data Architecture

2.1 Dataset Generation

A synthetic dataset was generated using make_classification from sklearn.datasets to simulate real-world student placement scenarios.

- **Total Samples**: 1000 students
- **Features**: 20 (representing CGPA, internships, projects, backlogs, etc.)
- **Class Distribution**:
    - 90% placed (majority class)
    - 10% unplaced (minority class)

This imbalance reflects real-world placement trends, where most students get placed, making the minority class (at-risk students) critically important.

2.2 Train-Test Split

The dataset was split into:

- **Training Set**: 80%
- **Testing Set**: 20%
- **Stratification Applied**: Ensures class distribution consistency across splits

2.3 Feature Scaling

Feature scaling was performed using **StandardScaler**:

- The scaler was **fit only on training data**

- Applied to both training and testing sets

This avoids **data leakage**, ensuring the model does not gain prior knowledge of the test distribution.

3. Baseline Model and Metric Analysis

3.1 Baseline Model

A default **RandomForestClassifier** was trained on the processed data.

Evaluation Metrics:

- **Accuracy**: Measures overall correctness
- **F1-Score**: Harmonic mean of precision and recall (critical for imbalanced data)

3.2 Observations

- The baseline model typically shows **high accuracy** due to class imbalance.
- However, **F1-score is lower**, indicating poor performance on minority class (at-risk students).

3.3 The Metric Trap

Accuracy can be misleading in imbalanced datasets. For example:

- If the model predicts all students as "placed," it achieves ~90% accuracy.
- However, it completely fails to identify at-risk students.

Thus, **F1-score is the more appropriate metric**, as it balances false positives and false negatives.

4. Hyperparameter Optimization

4.1 Grid Search (Accuracy Optimization)

GridSearchCV was used to exhaustively search over:

- n_estimators: [50, 100, 200]
- max_depth: [None, 10, 20]
- min_samples_split: [2, 5, 10]

Objective: Maximize **accuracy**

4.2 Grid Search (F1 Optimization)

The same parameter grid was used, but optimized for:

- **F1-score**

4.3 Analysis: Why Best Models Differ

The best model changes because:

- **Accuracy Optimization**:
    - Favors majority class predictions
    - Ignores minority class errors
    - Produces biased models
- **F1 Optimization**:
    - Balances precision and recall
    - Focuses on correctly identifying at-risk students
    - Produces more practically useful models

Thus, the choice of metric directly influences model behavior and parameter selection.

5. Efficiency Comparison

5.1 Grid Search Performance

- Performs **exhaustive search**
- Evaluates all parameter combinations
- Computationally expensive

5.2 Randomized Search

RandomizedSearchCV was implemented with:

- n_iter = 20
- Wider parameter range:
    - n_estimators: 10–500
    - max_depth: None, 10, 20, 30
    - min_samples_split: 2, 5, 10, 15

Advantages:

- Faster execution
- Explores broader search space
- Often finds near-optimal solutions

6. Comparative Analysis

Method	Optimization Metric	Time Taken	F1 Score	Key Insight
Baseline Model	None	Very Low	Moderate	No tuning
Grid Search (Accuracy)	Accuracy	High	Lower	Biased toward majority
Grid Search (F1)	F1 Score	High	High	Best balanced model
Randomized Search	F1 Score	Medium	Near High	Efficient alternative

Key Findings:

- **Grid Search (F1)** yields the best performance but is time-intensive.
- **Randomized Search** achieves comparable results with significantly less computation.
- **Grid Search (Accuracy)** produces misleading results in imbalanced scenarios.

7. Conclusion

This study demonstrates that:

1. **Metric selection is critical** in imbalanced classification problems.
2. **Accuracy alone is insufficient** for evaluating models in real-world scenarios.
3. **F1-score provides a more reliable evaluation**, especially when detecting minority classes.
4. **Randomized Search offers an efficient alternative** to exhaustive Grid Search with minimal performance trade-off.

For deployment in a university placement system:

- Models should be optimized using **F1-score**
- **Randomized Search** is recommended for scalability
- Focus should remain on **identifying at-risk students early**, rather than maximizing overall accuracy

8. Future Work

- Incorporation of real student datasets
- Use of advanced techniques like **SMOTE** for imbalance handling
- Evaluation using additional metrics (ROC-AUC, Precision-Recall curve)
- Deployment as a dashboard for placement officers